SIMATS ENGINEERING

ASSESSMENT TOOL 2

R.Poornima siva Priya Reg:192425181

COURSE NAME: Deep Learning COURSE CODE: MLA047

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Answer

Given:

Training MSE = 0.85

Testing MSE = 5.20

The training error is **very low**, while the testing error is **much higher**. This indicates that the neural network performs well on the training data but poorly on unseen test data.

Possible Reasons

1. Overfitting (Main Reason)

The neural network has memorized the training data instead of learning general patterns.

- Training data → Very accurate (MSE = 0.85)
- Test data → High error (MSE = 5.20)

Example:
Suppose a model is trained using the marks of **100 students**.

Training Data:

The model predicts almost perfectly.

But on new students:

The prediction error becomes large, increasing the test MSE.

2. Small Training Dataset

If only a small number of student records are used for training, the model cannot learn all possible patterns.

Example:
Training with only **80 students** instead of **2000 students**.

The model learns only limited information and performs poorly on new data.

3. Complex Neural Network

Using too many hidden layers or neurons makes the model unnecessarily complex.

Example:

- 8 hidden layers
- 500 neurons

Such a network may memorize training data rather than generalize.

4. Noise in Training Data

Incorrect or inconsistent data can affect learning.

Example:

The second record may be incorrect, confusing the model.

5. Data Distribution Difference

The training and testing datasets may have different characteristics.

Example:

Training data:

- Scores between **40–80**

Testing data:

- Scores between **85–100**

The model has not learned these higher-score patterns.

6. Lack of Proper Regularization

Without regularization, the model may overfit.

Regularization techniques help prevent the model from memorizing training data.

Corrective Measures

1. Increase Training Data

Use a larger dataset so the neural network learns more general patterns.

Example:
Increase from **100 students** to **5000 students**.

2. Apply Regularization

Use:

- L1 Regularization
- L2 Regularization

These reduce overfitting by limiting very large weights.

3. Use Dropout

Dropout randomly disables some neurons during training.

Example:

- Dropout = **0.5**
- 50% of neurons are temporarily ignored in each training step.

This improves generalization.

4. Early Stopping

Stop training when the validation error starts increasing.

This prevents the network from over-learning the training data.

5. Reduce Model Complexity

Instead of:

- 8 hidden layers
- 500 neurons

Use:

- 2–3 hidden layers
- 64–128 neurons

A simpler model is less likely to overfit.

6. Data Preprocessing

- Remove incorrect records.
- Handle missing values.
- Normalize or standardize features.

This improves the quality of the input data.

7. Cross-Validation

Use **k-fold cross-validation** to evaluate the model on different subsets of the data. This provides a more reliable estimate of model performance and helps detect overfitting.